

Bonus - Second-Order Predictor

Model

The second-order predictor is defined as:

$$\hat{x}[n] = a_1 x[n-1] + a_2 x[n-2]$$

Using the same sequence from Part A: $x[n] = \{10, 12, 15, 17, 18, 20, 21, 23\}$, $n = 0, \dots, 7$

Deriving the Optimum Coefficients

To minimize the squared prediction error $\sum e^2[n]$, the normal equations are solved:

$$a_1 \sum x^2[n-1] + a_2 \sum x[n-1]x[n-2] = \sum x[n]x[n-1]$$

$$a_1 \sum x[n-1]x[n-2] + a_2 \sum x^2[n-2] = \sum x[n]x[n-2]$$

Summation Table

Quantity	Value
$\sum x^2[n-1]$	1823
$\sum x^2[n-2]$	1482
$\sum x[n-1]x[n-2]$	1641
$\sum x[n]x[n-1]$	2004
$\sum x[n]x[n-2]$	1802

This gives the system:

$$1823 a_1 + 1641 a_2 = 2004$$

$$1641 a_1 + 1482 a_2 = 1802$$

Solving: $a_1 \approx 1.4589$, $a_2 \approx -0.3995$

Prediction Error and MSE

n	x[n]	$\hat{x}[n]$	e[n]
2	15	13.51	1.49
3	17	17.09	-0.09
4	18	18.81	-0.81
5	20	19.47	0.53
6	21	21.99	-0.99
7	23	22.65	0.36

$$\text{MSE}_2 \approx 0.7097$$

Comparison with First-Order Optimum Predictor

Using $a_{\text{opt}} = 1.1045$ from Part C, restricted to the same range ($n = 3:N$) for a fair comparison:

$$\text{MSE}_1 \approx 0.8465$$

Since $\text{MSE}_2 < \text{MSE}_1$, the second-order predictor achieves a lower prediction error than the optimum first-order predictor.

MATLAB Verification

The MATLAB output confirming the calculated coefficients and MSE values is shown below.

```
a_opt for first order predictor:
    1.1045

a1 = 1.4589 , a2 = -0.3995
MSE first order (n=3:N) = 0.8465
MSE second order = 0.7097
```

Figure 1: MATLAB command window output showing a_{opt} , a_1 , a_2 , and MSE values.

Plots

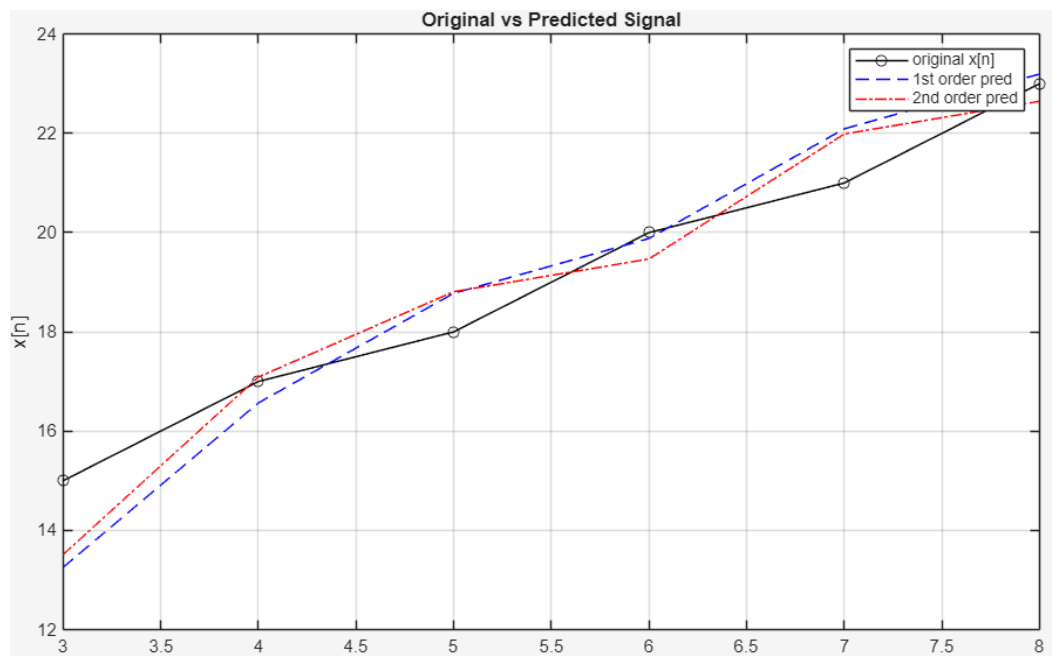


Figure 2: Original signal compared with first-order and second-order predictions.

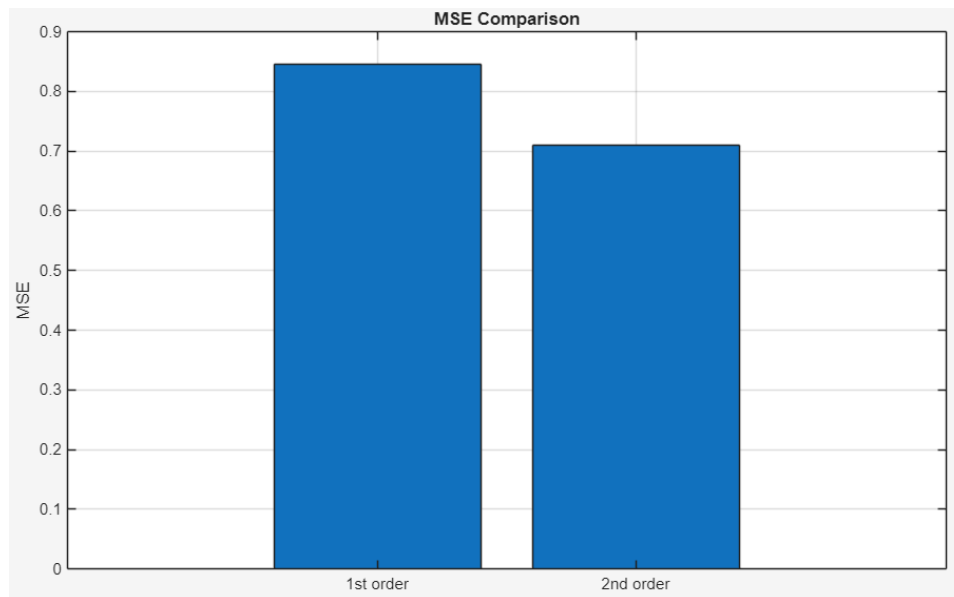


Figure 3: MSE comparison between first-order and second-order predictors.

Discussion and Conclusion

As shown in Figure 3, the second-order predictor achieves a lower MSE (≈ 0.71) compared to the optimum first-order predictor (≈ 0.85), an improvement of roughly 16%. This confirms that using two previous samples allows the predictor to exploit additional correlation in the signal.

However, increasing the predictor order is not always beneficial. Higher-order predictors generally reduce MSE further, but the improvement shows diminishing returns as the order increases, while each additional order requires an extra coefficient and additional multiply-accumulate operations per sample, which is costly in real-time systems such as speech coding. More coefficients also increase sensitivity to estimation errors, especially with limited or noisy data, and raise the risk of overfitting, and they must be stored and computed while additional past samples are kept in memory, increasing memory usage and delay. Taken together, increasing the predictor order improves performance only up to a point, governed by how much correlation actually exists between distant samples; beyond that point, the marginal reduction in MSE becomes small relative to the added complexity and sensitivity, so in practice, low-order predictors (order 2-4) are typically used as a practical trade-off.